Eight Queens Problem

```
def eight_queens(n=8):
    """Backtracking; returns (first_solution, total_solutions)."""
    solutions = []
    cols = []

    def safe(row, col):
        return all(c != col and abs(c - col) != row - r for r, c in enumerate(cols))

    def place(row):
        if row == n:
            solutions.append(cols[:])
            return
        for col in range(n):
            if safe(row, col):
                cols.append(col)
                place(row + 1)
                cols.pop()

    place(0)
    return solutions[0], len(solutions)


def program1():
    print("\n=== 1. 8 QUEENS ===")
    sol, total = eight_queens(8)
    print("Total solutions:", total)
    print("First solution (column of queen in each row):", sol)
    for c in sol:
        print(" ".join("Q" if i == c else "." for i in range(8)))


if __name__ == "__main__":
    program1()
```